Combinatorics of Saddle Points in Deep Linear Networks

Guillaume Corlouer

Abstract

Contents

1 Introduction: Informal statement of the main theorem **1**

2 Related works **2**

3 Setup and formal statement **2**

4 Geometry: Rank pattern stratification **2**

5 Stratification of the cyclic variety with multiplicity patterns **4**
5.1 Essentials of Quiver representation theory 4
5.2 Quiver representation of the cyclic variety 4
5.3 Specialization to fixed closing arrow 5

6 From multiplicity patterns to rank patterns **5**

7 Stratification of the critical loci **5**

8 Discussion **5**

9 Appendix **5**
9.1 Detailed proof of Proposition 4.1 5
9.2 Detailed proof of Proposition 4.2 7

10 Deprecated **9**

1 Introduction: Informal statement of the main theorem

Flow of the paper: The main theorem will be the stratification of the critical loci via the stratification of the cyclic variety and its intersection wit the specialization. We introduce notation, we state the main theorem (without any quiver representation theory simply in terms of the rank patterns).

2 Related works

3 Setup and formal statement

The critical loci of rank r critical points in the loss landscape of a deep linear network admits a stratification parametrized by the rank of the critical points, rank patterns and a subset S of singular vectors of the teacher matrix M.

Definition 3.1 (Determinantal, cyclic, and critical loci)**.** Define

$$\Sigma_d^r := \{\theta \in \Omega : \operatorname{rank}(\mu(\theta)) = r\} \quad \textit{Determinantal variety},$$

$$\Gamma_d := \{(\theta, W_0) \in \Omega \times \mathrm{Mat}_{d_0 \times d_L} \mid W_{<\ell} W_0 W_{>\ell} = 0 \text{ for all } 1 \le \ell \le L\}, \quad \textit{Cyclic variety}$$

Let $\iota_S : \Omega \longrightarrow \Omega \times \mathrm{Mat}_{d_0 \times d_L}$ be an inclusion map such that $\iota_S(\theta) := (\theta, \Sigma_{XY} P_Q)$ and define:

$$\Gamma_d^r := \Gamma_d \cap (\Sigma_d^r \times \mathrm{Mat}_{d_0 \times d_L}), \qquad \Gamma_d^S := \iota_S^{-1}(\Gamma_d^r)$$

Let S be a subset of left-singular vectors of M, define the S-critical loci:

$$\mathcal{C}_d^S = \{\theta \in \Omega | \ \nabla_\theta L(\theta) = 0; \mu(\theta) = P_S W^\star\}$$

Definition 3.2. For $\phi = (\theta, W_0) \in \Gamma_d$ we define its *cyclic rank pattern* to be the array

$$r(\phi) = (r_{ij}(\phi))_{0 \le i,j \le L} \quad \text{where} \quad r_{ij}(\phi) := \operatorname{rank}\big(W_j ... W_i\big) \quad \text{for} \quad \text{and} \quad r_{ii}(\phi) = d_i \text{and} \quad r_{i+1i}(\phi) = \operatorname{rank}\big(W_i\big)$$

Theorem 3.3. Define the strata $\mathcal{O}_\rho^S := \mathcal{O}_\rho \cap \Gamma_{d,S}^r$ where $\mathcal{O}_\rho$ is a G_d-orbit of the cyclic variety Γ_d^r. Assuming that the data is non degenerate and the input covariance matrix is invertible then the loci of critical points of rank $r < r_{\max}$ admits the following stratification parameterized by subset S of singular vectors of the teacher with r elements and admissible cyclic rank patterns $\rho \in R_d^{\mathrm{orb}}$:

$$\mathcal{C}_d^r = \bigsqcup_{|S|=r} \mathcal{C}_d^S = \bigsqcup_{|S|=r} \bigsqcup_{\substack{\rho \in R_d^{\mathrm{orb}} \\ \operatorname{rank}(\rho) = \operatorname{rank}(\Sigma_{XY} P_Q)}} \mathcal{O}_\rho^S$$

4 Geometry: Rank pattern stratification

The purpose of this section is to relate the S-critical locus to the specialization of the cyclic variety obtained by fixing the closing arrow to be $\Sigma_{XY} P_Q$. Recall that

$$\Gamma_d^S = \iota_S^{-1}(\Gamma_d^r), \qquad \iota_S(\theta) = (\theta, \Sigma_{XY} P_Q).$$

Thus $\theta \in \Gamma_d^S$ precisely when

$$(\theta, \Sigma_{XY} P_Q) \in \Gamma_d^r.$$

Proposition 4.1 (The S-critical locus as a product fiber)**.** Assume that Σ_X is invertible, that

$$W^* = \Sigma_{YX} \Sigma_X^{-1},$$

and that $\operatorname{rank}(P_S W^*) = r$. Then, as subsets of Ω,

$$\mathcal{C}_d^S = \left\{\theta \in \Gamma_d^S \ \middle| \ \mu(\theta) = P_S W^*\right\}.$$

Equivalently,

$$\mathcal{C}_d^S = \Gamma_d^S \cap \mu^{-1}(P_S W^*) = \iota_S^{-1}(\Gamma_d^r) \cap \mu^{-1}(P_S W^*).$$

Proof. The point is that the gradient equations with $\mu(\theta) = P_S W^*$ are exactly the cyclic equations with closing arrow $\Sigma_{XY} P_Q$. Indeed,

$$\Sigma_X (P_S W^*)^\top - \Sigma_{XY} = -\Sigma_{XY} P_Q.$$

Therefore

$$\big(\nabla_{W_\ell} \mathcal{L}(\theta)\big)^\top = -2 W_{<\ell} \Sigma_{XY} P_Q W_{>\ell}.$$

Thus the partial gradients vanish for all ℓ if and only if

$$W_{<\ell} \Sigma_{XY} P_Q W_{>\ell} = 0 \qquad \text{for all } 1 \le \ell \le L.$$

Together with $\mu(\theta) = P_S W^*$ and $\operatorname{rank}(P_S W^*) = r$, this is exactly the condition that $\theta \in \Gamma_d^S \cap \mu^{-1}(P_S W^*)$. See Appendix 9.1 for the full verification. □

Now assume that the rank r cyclic variety admits a finite cyclic rank-pattern decomposition

$$\Gamma_d^r = \bigsqcup_{\rho \in R} O_\rho^r,$$

where each O_ρ^r is locally closed in Γ_d^r. We assume that the rank-pattern pieces satisfy the frontier inclusion

$$\overline{O_\rho^r}^{\Gamma_d^r} \subseteq \bigcup_{\rho' \le \rho} O_{\rho'}^r,$$

where $\rho' \le \rho$ means that

$$\rho'_{ij} \le \rho_{ij} \qquad \text{for all } i, j.$$

This assumption is implied, for example, by the stronger equality

$$\overline{O_\rho^r}^{\Gamma_d^r} = \bigcup_{\rho' \le \rho} O_{\rho'}^r.$$

The decomposition of Γ_d^r induces a decomposition of the S-critical locus by pulling back along ι_S and then imposing the product constraint.

Proposition 4.2 (Induced cyclic rank-pattern decomposition of the critical locus)**.** Define

$$\mathcal{C}_\rho^S := \iota_S^{-1}(O_\rho^r) \cap \mu^{-1}(P_S W^*).$$

Then, after discarding empty pieces,

$$\mathcal{C}_d^S = \bigsqcup_{\rho \in R} \mathcal{C}_\rho^S.$$

Each $\mathcal{C}_\rho^S$ is locally closed in $\mathcal{C}_d^S$. Moreover, the relative closures satisfy

$$\overline{\mathcal{C}_\rho^S}^{\mathcal{C}_d^S} \subseteq \bigcup_{\rho' \le \rho} \mathcal{C}_{\rho'}^S.$$

Proof. By Proposition 4.1,

$$\mathcal{C}_d^S = \iota_S^{-1}(\Gamma_d^r) \cap \mu^{-1}(P_S W^*).$$

Since the O_ρ^r form a disjoint decomposition of Γ_d^r, their pullbacks along ι_S, intersected with $\mu^{-1}(P_S W^*)$, form a disjoint decomposition of $\mathcal{C}_d^S$. Local closedness follows because preimages of locally closed sets under continuous maps are locally closed, and $\mu^{-1}(P_S W^*)$ is closed.

For the frontier statement, use the elementary topological inclusion

$$\overline{f^{-1}(A)} \subseteq f^{-1}(\overline{A})$$

for continuous f, applied to the restriction of ι_S to $\mathcal{C}_d^S$. Together with the frontier inclusion for O_ρ^r, this gives

$$\overline{\mathcal{C}_\rho^S}^{\mathcal{C}_d^S} \subseteq \bigcup_{\rho' \leq \rho} \mathcal{C}_{\rho'}^S.$$

The full argument is written in Appendix 9.2. □

Thus the S-critical locus inherits a coarse decomposition by cyclic rank patterns. The closure statement is deliberately one-sided: along degenerations inside $\mathcal{C}_d^S$, cyclic ranks can only decrease. We do not claim that every lower cyclic rank pattern appearing in $\mathcal{C}_d^S$ is reached as a degeneration inside the fixed-S critical locus. We see that we can stratify the critical locus by stratifying the cyclic variety. In the following sections we will show how to obtain such stratification using quiver representation theory.

5 Stratification of the cyclic variety with multiplicity patterns

Goal: Stratify the cyclic variety with cyclic rank patterns.

5.1 Essentials of Quiver representation theory

A quiver is a graph (I, E) with vertices I, edges E and an orientation. A quiver representation assigns a vector space to $V_i \cong \mathbb{R}^{d_i}$ and a linear map $\phi : V_i \to V_{i+1}$ with vector dimension d_i.

5.2 Quiver representation of the cyclic variety

The cyclic variety is equivalent to a representation of a cyclic quiver with finite relation.

Definition 5.1. Define the cyclic quiver $Q = C_{L+1}$ with vertices $l \in \{0, ..., L\}$ and edges $a_l : l - 1 \longrightarrow l \bmod L + 1$ with $a_0 : L \longrightarrow 0$ Define the paths of length L for $l \in \{1, ..., L\}$:

$$p_l := a_{l-1} ... a_1 a_0 a_L ... a_{l+1} \bmod L + 1$$

.

Define the bounded path algebra with relations by quotienting the path algebra kQ by imposing that all paths of length L going through the closing arrow a_0 vanish.

Definition 5.2. Consider the path algebra kQ, consider the ideal:

$$I = \langle p_0, ..., p_L \rangle$$

5.3 Specialization to fixed closing arrow

6 From multiplicity patterns to rank patterns

7 Stratification of the critical loci

8 Discussion

9 Appendix

9.1 Detailed proof of Proposition 4.1

We prove the equality

$$\mathcal{C}_d^S = \Gamma_d^S \cap \mu^{-1}(P_S W^*) = \iota_S^{-1}(\Gamma_d^r) \cap \mu^{-1}(P_S W^*)$$

as an equality of subsets of Ω.

Recall that

$$\iota_S(\theta) = (\theta, \Sigma_{XY} P_Q),$$

and therefore

$$\Gamma_d^S = \iota_S^{-1}(\Gamma_d^r).$$

Thus $\theta \in \Gamma_d^S$ means that

$$(\theta, \Sigma_{XY} P_Q) \in \Gamma_d^r.$$

Equivalently,

$$\operatorname{rank}(\mu(\theta)) = r$$

and

$$W_{<\ell} \Sigma_{XY} P_Q W_{>\ell} = 0 \qquad \text{for all } 1 \le \ell \le L.$$

We first prove

$$\mathcal{C}_d^S \subseteq \Gamma_d^S \cap \mu^{-1}(P_S W^*).$$

Let $\theta \in \mathcal{C}_d^S$. By definition of the S-critical locus,

$$\nabla_\theta \mathcal{L}(\theta) = 0 \qquad \text{and} \qquad \mu(\theta) = P_S W^*.$$

Since $\operatorname{rank}(P_S W^*) = r$, it follows that

$$\operatorname{rank}(\mu(\theta)) = r.$$

It remains to show the cyclic equations with closing arrow $\Sigma_{XY} P_Q$.

For each layer ℓ, the transpose-gradient formula is

$$\big(\nabla_{W_\ell} \mathcal{L}(\theta)\big)^\top = 2 W_{<\ell} \big(\Sigma_X \mu(\theta)^\top - \Sigma_{XY}\big) W_{>\ell}.$$

Using $\mu(\theta) = P_S W^*$, we get

$$\Sigma_X \mu(\theta)^\top - \Sigma_{XY} = \Sigma_X (P_S W^*)^\top - \Sigma_{XY}.$$

Since P_S is an orthogonal projector, $P_S^\top = P_S$, and therefore

$$(P_S W^*)^\top = (W^*)^\top P_S.$$

Moreover, by the definition

$$W^* = \Sigma_{YX}\Sigma_X^{-1},$$

and by symmetry of Σ_X, we have

$$\Sigma_X(W^*)^\top = \Sigma_X(\Sigma_X^{-1})^\top\Sigma_{YX}^\top = \Sigma_X\Sigma_X^{-1}\Sigma_{XY} = \Sigma_{XY}.$$

Hence

$$\Sigma_X(P_SW^*)^\top - \Sigma_{XY} = \Sigma_X(W^*)^\top P_S - \Sigma_{XY} = \Sigma_{XY}P_S - \Sigma_{XY}.$$

Since $P_Q = I - P_S$, this becomes

$$\Sigma_X(P_SW^*)^\top - \Sigma_{XY} = -\Sigma_{XY}P_Q.$$

Substituting into the gradient formula gives

$$\big(\nabla_{W_\ell}\mathcal{L}(\theta)\big)^\top = -2W_{<\ell}\Sigma_{XY}P_QW_{>\ell}.$$

Because $\theta \in \mathcal{C}_d^S$, every partial gradient vanishes. Therefore

$$W_{<\ell}\Sigma_{XY}P_QW_{>\ell} = 0 \qquad \text{for all } 1 \le \ell \le L.$$

Thus

$$(\theta, \Sigma_{XY}P_Q) \in \Gamma_d^r,$$

or equivalently,

$$\theta \in \Gamma_d^S.$$

Since also $\mu(\theta) = P_SW^*$, we have

$$\theta \in \Gamma_d^S \cap \mu^{-1}(P_SW^*).$$

Conversely, suppose that

$$\theta \in \Gamma_d^S \cap \mu^{-1}(P_SW^*).$$

Then

$$\mu(\theta) = P_SW^*$$

and, because $\theta \in \Gamma_d^S$,

$$W_{<\ell}\Sigma_{XY}P_QW_{>\ell} = 0 \qquad \text{for all } 1 \le \ell \le L.$$

As above,

$$\Sigma_X\mu(\theta)^\top - \Sigma_{XY} = -\Sigma_{XY}P_Q.$$

Therefore, for every ℓ,

$$\big(\nabla_{W_\ell}\mathcal{L}(\theta)\big)^\top = -2W_{<\ell}\Sigma_{XY}P_QW_{>\ell} = 0.$$

Hence

$$\nabla_{W_\ell}\mathcal{L}(\theta) = 0 \qquad \text{for every } 1 \le \ell \le L.$$

So

$$\nabla_\theta\mathcal{L}(\theta) = 0.$$

Together with $\mu(\theta) = P_SW^*$, this gives

$$\theta \in \mathcal{C}_d^S.$$

This proves the reverse inclusion and therefore the proposition.

9.2 Detailed proof of Proposition 4.2

Assume that

$$\Gamma_d^r = \bigsqcup_{\rho \in R} O_\rho^r$$

is a finite decomposition into locally closed subsets of Γ_d^r, and that the pieces satisfy

$$\overline{O_\rho^r}^{\Gamma_d^r} \subseteq \bigcup_{\rho' \leq \rho} O_{\rho'}^r.$$

For each $\rho \in R$, define

$$\mathcal{C}_\rho^S := \iota_S^{-1}(O_\rho^r) \cap \mu^{-1}(P_S W^*).$$

We first prove that the pieces $\mathcal{C}_\rho^S$ give a disjoint decomposition of $\mathcal{C}_d^S$. By Proposition 4.1,

$$\mathcal{C}_d^S = \iota_S^{-1}(\Gamma_d^r) \cap \mu^{-1}(P_S W^*).$$

Since

$$\Gamma_d^r = \bigsqcup_{\rho \in R} O_\rho^r,$$

we have

$$\iota_S^{-1}(\Gamma_d^r) = \iota_S^{-1}\left(\bigsqcup_{\rho \in R} O_\rho^r\right).$$

Preimages commute with arbitrary unions, and preimages of disjoint sets are disjoint. Hence

$$\iota_S^{-1}(\Gamma_d^r) = \bigsqcup_{\rho \in R} \iota_S^{-1}(O_\rho^r).$$

Intersecting both sides with $\mu^{-1}(P_S W^*)$ gives

$$\mathcal{C}_d^S = \left(\bigsqcup_{\rho \in R} \iota_S^{-1}(O_\rho^r)\right) \cap \mu^{-1}(P_S W^*).$$

Therefore

$$\mathcal{C}_d^S = \bigsqcup_{\rho \in R} \left(\iota_S^{-1}(O_\rho^r) \cap \mu^{-1}(P_S W^*)\right) = \bigsqcup_{\rho \in R} \mathcal{C}_\rho^S.$$

After discarding the empty terms, this is the claimed decomposition.

Next we prove local closedness. Each O_ρ^r is locally closed in Γ_d^r. Since Γ_d^r is a locally closed subset of $\Omega \times \mathrm{Mat}_{d_0 \times d_L}$, each O_ρ^r is locally closed in $\Omega \times \mathrm{Mat}_{d_0 \times d_L}$. The map

$$\iota_S : \Omega \longrightarrow \Omega \times \mathrm{Mat}_{d_0 \times d_L}$$

is continuous, so

$$\iota_S^{-1}(O_\rho^r)$$

is locally closed in Ω. The map $\mu : \Omega \to \mathrm{Mat}_{d_L \times d_0}$ is regular, hence continuous, and the point $\{P_S W^*\}$ is closed. Therefore

$$\mu^{-1}(P_S W^*)$$

is closed in Ω. Consequently

$$\mathcal{C}_\rho^S = \iota_S^{-1}(O_\rho^r) \cap \mu^{-1}(P_S W^*)$$

is locally closed in Ω. Since

$$\mathcal{C}_\rho^S \subseteq \mathcal{C}_d^S,$$

it is also locally closed in $\mathcal{C}_d^S$ with the subspace topology.

It remains to prove the frontier inclusion

$$\overline{\mathcal{C}_\rho^S}^{\mathcal{C}_d^S} \subseteq \bigcup_{\rho' \leq \rho} \mathcal{C}_{\rho'}^S.$$

We use the following elementary fact. If $f : X \to Y$ is continuous and $A \subseteq Y$, then

$$\overline{f^{-1}(A)}^X \subseteq f^{-1}\left(\overline{A}^Y\right).$$

Indeed, $f^{-1}(\overline{A}^Y)$ is closed in X and contains $f^{-1}(A)$, so it contains the closure of $f^{-1}(A)$.

Apply this fact to the continuous map

$$\iota_S|_{\mathcal{C}_d^S} : \mathcal{C}_d^S \longrightarrow \Gamma_d^r.$$

This restriction is well-defined because, by Proposition 4.1,

$$\mathcal{C}_d^S \subseteq \Gamma_d^S = \iota_S^{-1}(\Gamma_d^r),$$

so $\iota_S(\theta) \in \Gamma_d^r$ for every $\theta \in \mathcal{C}_d^S$.

Now observe that

$$\mathcal{C}_\rho^S = (\iota_S|_{\mathcal{C}_d^S})^{-1}(O_\rho^r).$$

Therefore

$$\overline{\mathcal{C}_\rho^S}^{\mathcal{C}_d^S} = \overline{(\iota_S|_{\mathcal{C}_d^S})^{-1}(O_\rho^r)}^{\mathcal{C}_d^S} \subseteq (\iota_S|_{\mathcal{C}_d^S})^{-1}\left(\overline{O_\rho^r}^{\Gamma_d^r}\right).$$

Using the assumed frontier inclusion,

$$\overline{O_\rho^r}^{\Gamma_d^r} \subseteq \bigcup_{\rho' \leq \rho} O_{\rho'}^r,$$

we obtain

$$\overline{\mathcal{C}_\rho^S}^{\mathcal{C}_d^S} \subseteq (\iota_S|_{\mathcal{C}_d^S})^{-1}\left(\bigcup_{\rho' \leq \rho} O_{\rho'}^r\right).$$

Preimages commute with unions, so

$$(\iota_S|_{\mathcal{C}_d^S})^{-1}\left(\bigcup_{\rho' \leq \rho} O_{\rho'}^r\right) = \bigcup_{\rho' \leq \rho} (\iota_S|_{\mathcal{C}_d^S})^{-1}(O_{\rho'}^r).$$

But

$$(\iota_S|_{\mathcal{C}_d^S})^{-1}(O_{\rho'}^r) = \mathcal{C}_{\rho'}^S.$$

Hence

$$\overline{\mathcal{C}_\rho^S}^{\mathcal{C}_d^S} \subseteq \bigcup_{\rho' \leq \rho} \mathcal{C}_{\rho'}^S.$$

This proves the proposition.

The argument gives an inclusion rather than an equality. The possible failure of equality comes from the strictness of

$$\overline{f^{-1}(A)}^{X} \subseteq f^{-1}\left(\overline{A}^{Y}\right)$$

for a general continuous map f. In the present setting, a point in $\iota_S^{-1}(\overline{O_\rho^r}^{\Gamma_d^r})$ may be reachable as a degeneration of O_ρ^r in the full cyclic variety while not being reachable by a degeneration that remains inside the fixed specialization determined by ι_S. Therefore the natural statement for the S-critical locus is the one-sided frontier condition above.

10 Deprecated